

ANKIT KUMAR TIWARI

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SUMMARY

Dedicated Computer Science Engineering student at VIT Bhopal with a strong focus on Machine Learning and Deep Learning. Proficient in developing AI-driven solutions, including real-time computer vision systems and containerized ML pipelines. Active competitive programmer on Codeforces and LeetCode, passionate about solving complex algorithmic challenges and building scalable, data-centric applications.

EDUCATION

Vellore Institute of Technology, Bhopal

Expected Graduation: 2028

Bachelor of Technology in Computer Science and Engineering

Delhi Public School, Dagapur

Higher Secondary (Class 12th)

Marks: 88.4%

St. Francis' School, Jorethang

Secondary School (Class 10th)

Marks: 93.8%

SKILLS

Programming Languages: Python, C++, Java, JavaScript, SQL, C#, HTML, CSS

AI/ML & Data Science: OpenCV, YOLOv8, HuggingFace Transformers, PyKriging, Gemini LLM

Frameworks & Tech: FastAPI, React.js, Next.js 14, Node.js, ReactNative, MySQL

Tools: Docker Compose, Git, GitHub Actions, Blender, Unity

PROJECTS

AQI Live Monitor & Predictive Map

- Built an interactive heatmap dashboard with GPS tracking and EPA-standard color gradients.
- Applied Ordinary Kriging to interpolate sparse sensor data into a continuous AQI surface.
- Established a CI/CD pipeline with GitHub Actions to automate data fetching and deployment.

AI Vision Assist

- Developed a real-time YOLOv8-based obstacle detection system for visually impaired individuals.
- Engineered a spatial awareness algorithm to dynamically identify hazard directions (Left/Center/Right).
- Implemented a non-blocking voice alert mechanism using Python threading to prevent UI freezing.

AI-Powered Career Coaching System

- Built a hybrid AI pipeline using fine-tuned Transformers for career prediction and Gemini LLM for coaching.
- Orchestrated containerized microservices using Docker Compose with isolated networking.
- Developed a responsive Next.js 14 frontend with Pydantic-validated APIs and structured logging.

EXTRACURRICULARS & PARTICIPATIONS

- 6th Place, CodeVerse Hackathon, VIT Bhopal.
- Participant in Smart India Hackathon (SIH) and Dawn of the Code Hackathon.
- Active Member, LeetCode and Codeforces competitive programming communities.
- Interests: Cricket and Technical Problem Solving.